FINANCE RESEARCH / EQUITY SIGNALS

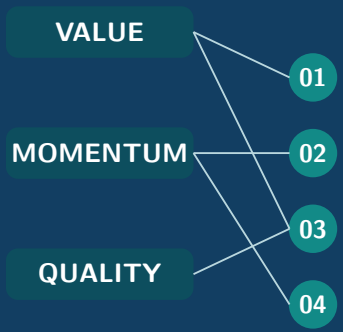
The Ruler Matters

When different signals lead to the same stocks

Mingyang Liu | **Imperial College London**

October 2026 / 20-minute research brief

ONE UNIVERSE / MULTIPLE SIGNA



STOCK-LEVEL EXPOSURES

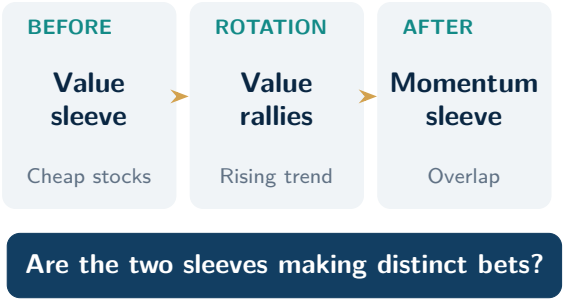
When value turns into momentum

FINANCIAL TIMES / MARKET REPORTING

Investors brace for “major shift” as momentum and value collide

Factor categories can move together

WHAT THE INVESTOR MUST ASK



Different factor names do not guarantee independent stock positions.

FT market reporting; flow schematic, not paper evidence.

One stock can carry several factor labels

HYPOTHETICAL SIX-STOCK UNIVERSE

STOCK	VALUE	MOMENTUM	QUALITY
Regional bank	✓	✓	
Energy producer	✓	✓	
Industrial supplier	✓		✓
Tech compounder		✓	✓
Retail franchise	✓		✓
Utility operator			✓

THE ISSUE

2

labels can apply to a stock

A factor label classifies; a stock exposure measures actual overlap.

All six labels and memberships are hypothetical; no empirical holdings.

More signals increase the need for a meaningful ruler

THE SIGNAL TOOLKIT KEEPS EXPANDING

36
CHARACTERISTICS



153
CHARACTERISTICS



EFFECTIVE STOCK-EXPOSURE DIRECTIONS

10.22 → **21.08**

ML forecasts	Conditional loadings	Portfolio targets

Effective direction count is participation dimension, not matrix rank or factor count.

Paper Figure 2: 225 formation cross sections, 2007–2025.

Where should we measure “difference”?

LENS 1 / SIGNAL RECIPES

What the analyst writes

Value $\times 1$ + Momentum $\times 0$

Value $\times 0$ + Momentum $\times 1$

Two distinct recipes

APPLY TO CURRENT STOCKS

STOCK UNIVERSE

$s_a = Za, \quad s_b = Zb$

LENS 2 / REAL POSITIONS

What the investor holds



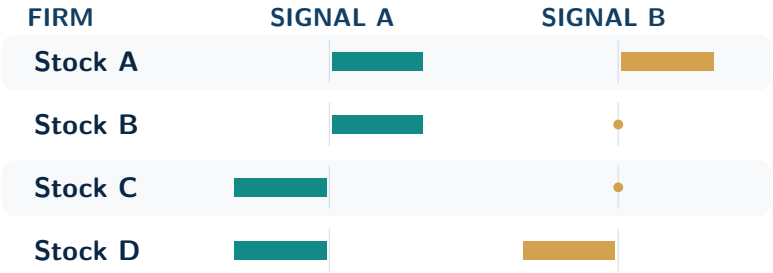
Shared longs and shorts

The stock cross section is where we must compare stock exposures.

Schematic linear signals mapped to exactly four hypothetical stocks.

Four stocks: different recipes, overlapping positions

WHAT EACH SIGNAL WOULD HOLD



LOOK AT THE STOCKS

A & B

same long on A
same short on D

$$a^\top b = 0 \qquad \text{BUT} \qquad \frac{(Za)^\top (Zb)}{4} = 0.50$$

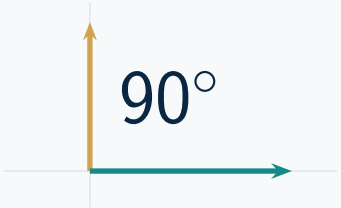
Coefficient orthogonality does not imply stock-exposure orthogonality.

Illustrative, centered exposures; calculations independently verified.

Why 90° on paper becomes 45° in stocks

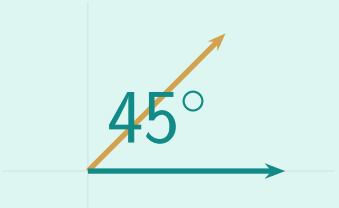
COEFFICIENT COORDINATES

Two perpendicular recipes



STOCK EXPOSURE SPACE

Two overlapping stock profiles



$$\cos\theta_{\text{stocks}} = \frac{0.50}{\sqrt{1.00 \times 0.50}} = \frac{1}{\sqrt{2}}$$

The same recipe directions acquire different angles when mapped onto stocks.

Exact angles in the verified four-stock toy; not return correlation.

The Gram matrix is the stock-space ruler

01 Describe
Coefficient directions

→

02 Apply
Positions in stocks

→

03 Compare
Actual overlap

EXACT IDENTITY

$$\frac{(Za)^\top (Zb)}{N} = a^\top Q b$$

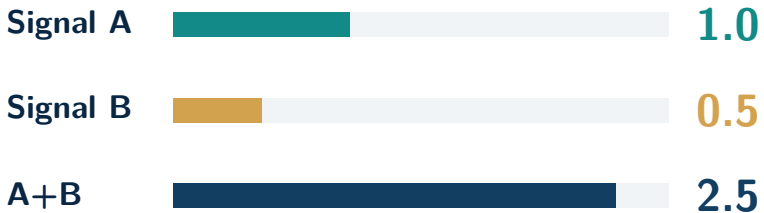
$$Q = \frac{Z^\top Z}{N}$$

This is an exposure ruler already present in IPCA, not a new factor estimator.

Liu manuscript, exact algebraic identity (2), (9).

The ruler reappears in exposure dispersion

VARIATION ACROSS THE FOUR STOCKS



CROSS-TERM

+1.0

from shared stock profiles

$$\mathrm{Var}_i[Z(a+b)] = 1.0 + 0.5 + 2(0.5) = 2.5$$

This is cross-sectional dispersion of loadings—not future portfolio-return variance.

Four-stock toy; cross-sectional exposure variance, NOT portfolio return risk.

Change feature units. Stock positions do not move.

ORIGINAL UNITS

$$\text{Score} = 1 \times \text{feature}$$

Same exposures in every stock



RE-EXPRESSED UNITS

$$\text{Score} = .01 \times (100 \times \text{feature})$$

Same exposures in every stock

INVARIANCE UNDER A CHANGE OF FEATURE COORDINATES

$$Z^* = ZD,\ a^* = D^{-1}a \ \Rightarrow\ (a^*)^\top Q^* b^* = a^\top Q b$$

The overlap should not change when only the units of the features change.

Illustrative transformation. Manuscript §IV.A and scale-invariance appendix.

Ruler I: what exposures do factors represent?

THE INVESTOR'S CHARACTERISTIC TARGET

Momentum tilt across today's stocks

↓

How much belongs to the factor-loading span?

PROFILE COVERAGE

Target stock profile



Covered by factor space | Residual

$$\text{Coverage} = \frac{a^\top QL(L^\top QL)^{-1}L^\top Qa}{a^\top Qa}$$

It is the entire loading span that matters, not arbitrary numbered latent factors.

Liu manuscript, eq. (10) and Figure 3: loading-span representation.

Better fit can mean weaker exposure coverage

COMPLETE RETURN FIT

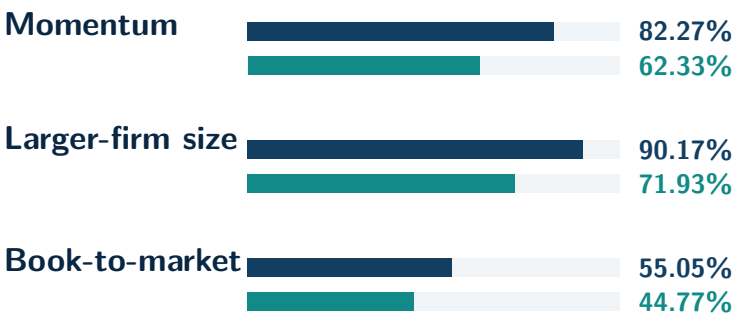
17.01%
36 features

↓

17.70%
153 features

+0.69 pp

SQUARED SHARE OF TARGET PROFILE IN JOINT LOADING SPAN

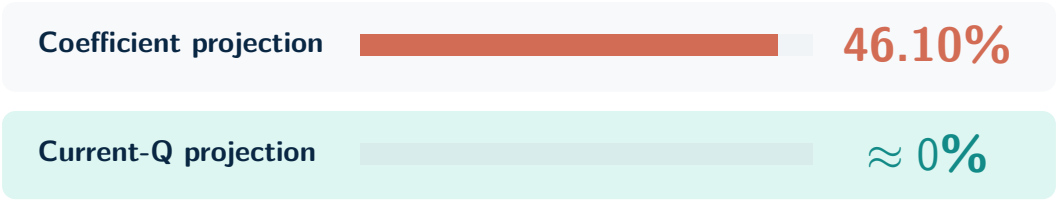


Fit: 599 months; coverage: 19 March origins.

Reported estimates; historical R^2 and rolling coverage use different windows.

Ruler II: is the hedge truly neutral in stocks?

STOCK-WEIGHT OVERLAP LEFT INSIDE THE FITTED LOADING SPAN



AFTER REMOVING LOADINGS, RESTORE THE SAME ORIGINAL CHARACTERISTIC TARGET

1. **Choose tilt** — Value or momentum
2. **Remove loadings** — Actual stock profiles
3. **Restore tilt** — Rescale holdings

46.10% is not a 46.10% reduction in portfolio-return variance.

Unrestricted 153-characteristic book-to-market; stock weights, not return variance.

Origin-Q vs Current-Q: today is not March

WHAT IS HELD FIXED?

Same annual loadings / current stocks / positive target; both portfolios form monthly



ORIGIN-Q

March stock cross section

$$Q_{\text{origin}} = \frac{Z_{\text{Mar}}^{\top} Z_{\text{Mar}}}{N_{\text{Mar}}}$$

CURRENT-Q

Each month's stock cross section

$$Q_{\text{current},t} = \frac{Z_t^{\top} Z_t}{N_t}$$

26.62–69.39% lower variance | 9/12 simultaneous intervals exclude zero

Different contemporaneous loading restrictions; this policy comparison is exploratory, not causal.

Exploratory positive-target policies; 216 common subsequent return months.

Measure overlap where stock positions live

Different features do not imply different stock exposures.

01 / INTERPRET

What does the model capture?

Compare a characteristic target with the factor-loading span.

02 / IMPLEMENT

What does the hedge remove?

Neutralize the loadings of the stocks actually held.

SCIENTIFIC BOUNDARIES

Exposure $\neq$ return covariance

Neutrality $\neq$ minimum risk

Current-Q result is empirical

Source: Liu (2026). Q measures stock-profile overlap, not return covariance.